Operating Systems

(Sistemas Operativos)

CPU Virtualization Mechanisms

University of Minho

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What will we learn?

CPU Management

- Although a computer has a limited number of CPUs, users get the impression that several programs (a lot more than the available CPUs) are running simultaneously
  - How does the OS provide this illusion?
  - How can users use the OS APIs to start and stop programs?
  - How does the OS choose which programs to run, switch, …?

- Let us start with the **mechanisms**!

CPU Virtualization

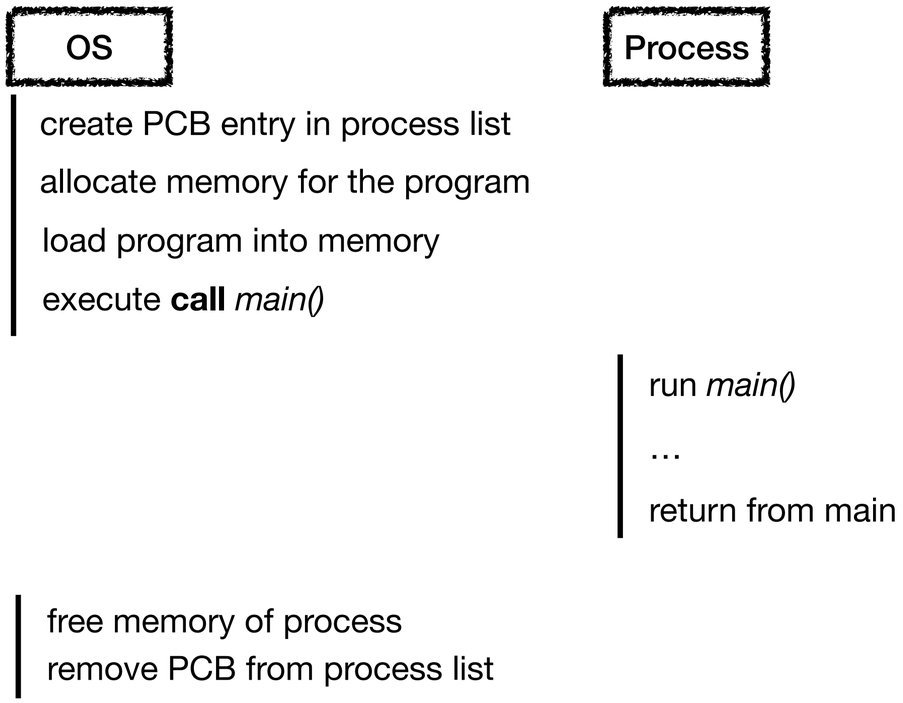
Challenges

- To virtualize the CPU, the OS must share it among many programs running seemingly at the same time
  - Basic idea: **Time-sharing** the CPU
    i.e., run a process for a little while, then another one, …
- Challenges
  - **Performance:** OS mechanisms cannot add excessive overhead to the system
  - **Control:** The OS must retain control over the CPU, avoiding processes from
    - Taking control of the CPU (e.g., infinite loops)
    - Accessing/corrupting information from other programs in memory or on disk

Limited Direct Execution

Basic Technique

- **Goal:** Run a program as fast as possible on the CPU (Performance)
  - **Solution:** The program runs natively on the hardware CPU (*direct execution*)
- **Issues:** If the program has full control over the CPU, how does the OS…
  - Ensure the process does not perform any unwanted operations?
  - Stop the process from running and switch it with another one?
- One needs to *limit the execution* of programs *(i.e.,* limit what they can do*)*



Modes of Execution

Restricted Operations

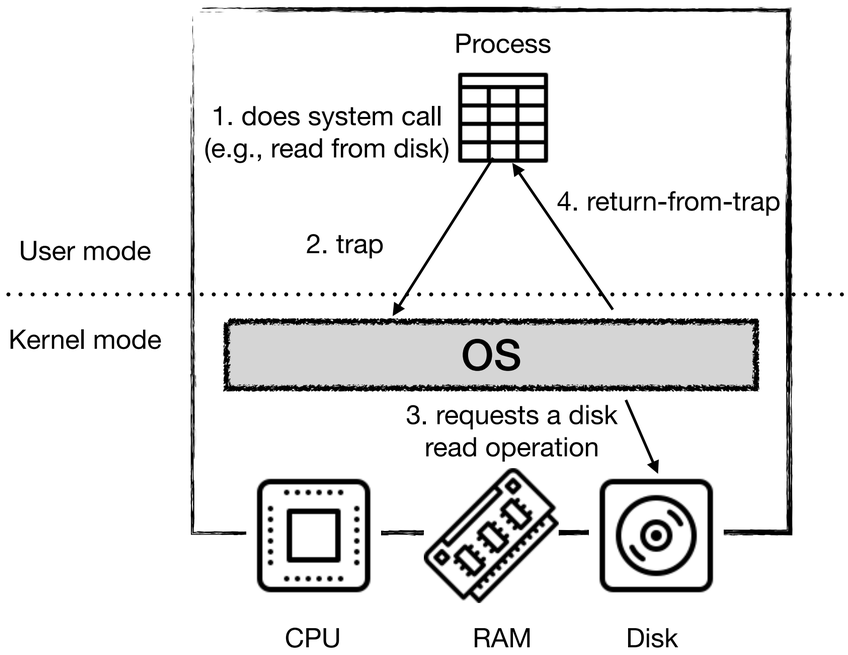
- User processes (*i.e.*, running programs) cannot freely do restricted operations (*e.g.*, I/O requests to disk, gain access to resources such as CPU or memory)
- The OS must mediate this interaction (with the aid of hardware)!
  The CPU provides two modes of execution[1]
  - **User mode:** Code running in the CPU cannot do restricted (privileged) operations. (e.g., a process in user mode cannot directly do disk I/O operations)
  - **Kernel mode:** Code executing in the CPU has full access to the hardware and can do any restricted operation. **The OS runs in this mode!**
- **Issues:** What if a user process attempts a privileged operation?
  - Remember the system calls?

[1] *A single bit, stored at the processor, indicates which mode the CPU is currently running in*

System Calls

Restricted Operations

- **System calls** allow the kernel (OS) to expose privileged operations to user processes (e.g., process creation, communication, memory allocation, I/O)
- To execute a system call, the process calls a special **trap** instruction, raising the privilege level to **kernel mode** and jumping into the OS code (**trap handler**)
- The OS can now perform privileged operations (on behalf of the user process)
- When finished, the OS calls a **return-from-trap** instruction that reduces the privilege level to **user mode** and resumes the execution of the process



Limited Direct Execution (Updated)

OS (Kernel Mode)

Hardware

Process (User Mode)

Limited Direct Execution (Updated)

OS (Kernel Mode)	Hardware	Process (User Mode)
create PCB entry / allocate and load memory **return-from-trap**		

Limited Direct Execution (Updated)

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	move to user mode jump to main	

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		run *main() …* *call system call* ***trap*** *into OS*

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handle trap do work of system call **return-from-trap**		
	move to user mode jump to PC after trap	

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		… *return from main* ***trap*** *(via* exit()*)*

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	move to kernel mode jump to trap handler	
handle trap do work of system call **return-from-trap**		
	move to user mode jump to PC after trap	
		… *return from main* ***trap*** *(via* exit()*)*
	move to kernel mode jump to trap handler	
free process memory / remove PCB		

Mode Switching

Traps and Interrupts

- **Two types of instructions** lead to transferring control from user to kernel mode
- A **trap** is generated by the current process running on a processor
  - The user program asks for an OS service (as we saw before)
    e.g., when the process issues a system call (in UNIX)
  - The user program performs an illegal action (an exception)
    e.g., division by zero, illegal memory access, …
- **Interrupts** are caused by devices and may not be related to the currently running process
  - When a device (disk, keyboard, network card) requires the OS's attention
    e.g., a disk read is ready, there is keyboard input to be read

Note: *Terminology changes depending on the system (x86, xv6). Some call trap to all instructions, other use the terms interchangeably…*

Interrupts

Why are interrupts useful?

- **Example:** When a process issues a read for a file (system call), the OS will ask the disk device for that specific content
  - The disk will require some time to bring that information to memory…
- The OS can periodically check (**polling**) the disk device for the request completion
  - If events take a long time to complete, then the OS is wasting precious CPU cycles that could be used to run other processes!
- **Interrupts** are generated by the devices asynchronously
  - **Benefit:** The OS is notified when events occur, leaving the CPU free for other processes
  - **Caution:** Interrupts are issued concurrently when other processes or the OS are running at the CPU

Limited Direct Execution (Interrupts)

OS (Kernel Mode)	Hardware	Process (User Mode)
create PCB entry / allocate and load memory **return-from-trap**		
	move to user mode jump to main	
		run *main()* *…*

Limited Direct Execution (Interrupts)

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create PCB entry / allocate and load memory **return-from-trap**		
	move to user mode jump to main	
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Limited Direct Execution (Interrupts)

OS (Kernel Mode)	Hardware	Process (User Mode)
create PCB entry / allocate and load memory **return-from-trap**		
	move to user mode jump to main	
		run *main()* *…*
	interrupt event (e.g., from disk) move to kernel mode jump to interrupt handler	
handle interruption do work **return-from-interrupt**		
	move to user mode jump to PC after interruption	
		… *return from main* ***trap*** *(via* exit()*)*

Traps and Interrupts

Process state

- Both **trap** and **interrupt** instructions are **hardware-assisted**
  - The OS cannot take the CPU from a process without aid
  - The CPU must interrupt the process, change to kernel mode, save some information, and jump to the OS code
- When a process is interrupted, the CPU must save a set of registers
  - Registers contain information (e.g., program counter) required by the CPU to resume the process execution when the OS calls a **return-from-trap/interrupt** instruction
  - Registers are stored, for example, in a per-process kernel stack

*On x86, for example, the processor will push the **program counter, flags, and a few other registers** onto its **kernel stack**; the return-from-trap will pop these values off the stack and resume execution of the user-mode program*

Limited Direct Execution (Registers)

OS (Kernel Mode)

Hardware

Process (User Mode)

Limited Direct Execution (Registers)

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create PCB / allocate and load memory **fill kernel stack with registers / PC** **return-from-trap**		

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	restore regs (from kernel stack) move to user mode jump to main	

Limited Direct Execution (Registers)

OS (Kernel Mode)

create PCB / allocate and load memory
fill kernel stack with registers / PC
return-from-trap

Hardware

restore regs (from kernel stack)
move to user mode
jump to main

Process (User Mode)

run *main() …*
*call system call / **trap** into OS*

Limited Direct Execution (Registers)

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create PCB / allocate and load memory **fill kernel stack with registers / PC** **return-from-trap**		
	restore regs (from kernel stack) move to user mode jump to main	
		run *main() …* *call system call / **trap** into OS*
	save regs (to kernel stack) move to kernel mode jump to trap handler	

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handle trap do work of system call **return-from-trap**		

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	restore regs (from kernel stack) move to user mode jump to PC after trap	
		… *return from main / **trap** (via* exit()*)*

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	save regs (to kernel stack) move to kernel mode jump to trap handler	
handle trap do work of system call **return-from-trap**		
	restore regs (from kernel stack) move to user mode jump to PC after trap	
		… *return from main / **trap** (via* exit()*)*

Switching Processes

The Conundrum

- Let's go back to our initial goal
  - **Time-sharing** the CPU, i.e., run a process for a little while, then another one…
  - This requires pausing one process and running another one
  - But we saw that if a process is running on the CPU, the OS is not running
- **Question:** Given the mechanisms previously studied, do you have any ideas on how the OS can share the CPU across several processes?
- We are getting closer to revealing the illusion created by the OS!

Cooperative Approach

Wait for System Calls

- When a process does a system call, the OS kernel regains control to
  - Handle the system call (e.g., ask the disk to read some content)
  - Do other tasks, such as scheduling another process!
- Actually, the OS can also regain control under device interruptions and other traps (exceptions)
- In a cooperative approach, the OS assumes the processes will
  - Issue system calls (e.g., block for doing disk I/O)
  - or voluntarily yield the CPU to the OS (the *yield* system call serves exactly this purpose)
- **Question:** What if we have malicious or buggy programs (e.g., infinite loops)…? Any ideas on how to solve it?

Non-Cooperative Approach

Timer Interrupt

- We need help from the hardware once again, namely from a **timer interrupt**
  - *E.g., e*very X milliseconds, an interrupt is generated by the hardware
  - The process is halted, a specific interrupt handler runs, and the OS gains control!
- The OS is responsible for
  - Defining the code for the timer interrupt handler
  - Setting the timer's interval (at boot time)
- **Remember:** when the interrupt occurs process state must be saved so the process can be resumed later…

Saving and Restoring Context

Context Switching

- When the OS regains control (cooperatively or non-cooperatively)
  - It must decide to keep running the same process or to run another one
  - This decision is made by the OS **scheduler**
- When the scheduler decides to switch process_A with process_B, a **context switch** is made
  - The OS must save information of process_A (e.g., program counter, memory limits, kernel stack pointer[1]) into its PCB_A structure (**Process Control Block** - check the *Processes slides*!)
    - This information is part of the **process list**, and it is managed by the OS! (Different from the per-process kernel stack managed by the hardware)
  - Switches contexts, by changing the CPU's stack pointer (a register) to the kernel stack of process_B

[1] Remember that the CPU stores essential information to resume a process in a per-process kernel stack. The PCB stores a pointer to the stack's address.

Limited Direct Execution (Context Switching)

OS (Kernel Mode)

Hardware

Processes (User Mode)

Limited Direct Execution (Context Switching)

OS (Kernel Mode)

Hardware

Processes (User Mode)

Process A
...

Limited Direct Execution (Context Switching)

OS (Kernel Mode)	Hardware	Processes (User Mode)
		Process A …
	timer interrupt **save regs (to kernel stack$_A$)** move to kernel mode jump to interrupt handler	

Limited Direct Execution (Context Switching)

OS (Kernel Mode)	Hardware	Processes (User Mode)
		Process A …
	timer interrupt **save regs (to kernel stack$_A$)** move to kernel mode jump to interrupt handler	
handle interrupt **call *switch()* routine** save PCB$_A$ restore regs from PCB$_B$ switch to kernel stack$_B$		

Limited Direct Execution (Context Switching)

OS (Kernel Mode)	Hardware	Processes (User Mode)
		Process A …
	timer interrupt **save regs (to kernel stack$_A$)** move to kernel mode jump to interrupt handler	
handle interrupt **call *switch()* routine** save PCB$_A$ restore regs from PCB$_B$ switch to kernel stack$_B$		
	restore regs (from kernel stack$_B$) move to user mode jump to Process$_B$ PC	

Limited Direct Execution (Context Switching)

OS (Kernel Mode)	Hardware	Processes (User Mode)
		Process A …
	timer interrupt **save regs (to kernel stack$_A$)** move to kernel mode jump to interrupt handler	
handle interrupt **call *switch()* routine** save PCB$_A$ restore regs from PCB$_B$ switch to kernel stack$_B$		
	restore regs (from kernel stack$_B$) move to user mode jump to Process$_B$ PC	
		Process B …

Performance

Context Switching

- In 1996, when running Linux 1.3.37 on a 200-MHz P6 CPU, **system calls took roughly 4 microseconds**, and a **context switch roughly 6 microseconds**
- Modern systems perform almost an order of magnitude better, with **sub-microsecond** results on systems with 2- or 3-GHz processors
- Whenever a system call to disk is made, for example, the OS will probably schedule another process to run
  - **Question:** Why?
  - **Question:** Do you see an impact of implementing a program that does repetitive system calls that take few microseconds?

More Information

- **Chapters 6** - Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau. **Operating Systems: Three Easy Pieces.** Arpaci-Dusseau Books, 2018.
- Avi Silberschatz, Peter Baer Galvin, Greg Gagne. **Operating System Concepts (10. ed)**. John Wiley & Sons, 2018.

Other Details

Limited Direct Execution (Trap Handler)

OS (Kernel Mode)	Hardware	Process (User Mode)
create PCB / allocate and load memory **fill kernel stack with registers / PC** **return-from-trap**		
	restore regs (from kernel stack) move to user mode jump to main	
		run *main() …* *call system call / **trap** into OS*
	save regs (to kernel stack) move to kernel mode jump to trap handler	
handle trap do work of system call **return-from-trap**		

How does the hardware knows what OS code it should run (it should jump into)?

Traps and Interrupts

Interrupt/Trap Table

- At boot time, the kernel (OS) sets up an **interrupt/trap table**
- The table tells the hardware what code to run (**interrupt/trap handler**) under exceptional events
  - What code should run when a hard disk interrupt happens?
  - Or when a keyboard interrupt occurs?
  - Or when a program makes a system call?
- The table maps interrupts and traps (e.g., identified by a number) to their corresponding handlers
  - It distinguishes interrupts from different **devices** (e.g., disk, keyboard) and **requests** (e.g, disk read, write)
  - It distinguishes between different **trap types** (e.g., exception, system call)

O	HANDLER SYSTEM CALL
1	HANDLER KEYBOARD…
2	HANDLER DISK…
…	…

Illustrative example
(not a real one!)

System Calls in Practice

Integration with the C library

- When a program calls a system call (*e.g.*, *fork()*, *open()*), a procedure call into the C library is made
- The library uses an agreed-upon calling convention with the OS kernel
  - **System call arguments** and **number** (identifier) are placed into well-known locations (e.g., stack, special registers)
- The library executes the corresponding trap instruction (system call)
  - The **interrupt/trap table** is used to check for the **system call trap handler**
  - When running the trap handler code, the OS can now validate the system call **number** and run the corresponding code
- This indirection provides **security**
  (safer than jumping directly to a memory address to run the code)

Questions?